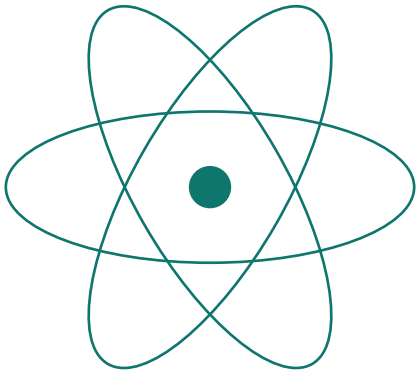
SPhOH

SAUDI PHYSICS OLYMPIAD HUB



NSMO Collection II

Olympiad Problem Collection

Olympiad Training • Advanced Mechanics & Mathematical Methods

SPhOH Team

FIRST EDITION • 2026

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About This Collection

This collection is part of the training resources developed by the **Saudi Physics Olympiad Hub (SPhOH)**. It serves as the second volume of the National Science and Mathematics Olympiad (NSMO) physics series.

The handout brings together three key areas of competitive physics:

1. **Dimensional Analysis** — Mathematical consistency, Buckingham Π theorem, and physical scaling laws.
2. **Kinematics II** — Advanced relative motion, non-inertial reference frames, constraint mechanics, and rotational kinematics.
3. **Dynamics I** — Newton's laws of motion, friction, multi-body coupled systems, and static equilibrium.

How to Use This Problem Set

Students are encouraged to attempt each problem independently before looking at its hint. A hint should be treated as a way to overcome a specific obstacle, not as a substitute for solving the problem.

Difficulty Levels

The problems are organized strictly by difficulty level within each section:

- **Level I** — Fundamental problems involving direct algebraic calculations and core physical laws.
- **Level II** — Multi-step problems combining moving frames, friction, non-inertial systems, and constraints.
- **Level III** — Challenging national problems requiring advanced analytical techniques.
- **Level IV** — Advanced competitive problems requiring deep physical insight and calculus.

Find more resources at: `https://devfhd.github.io/SPhOH/`.

Part I

Dimensional Analysis

CHAPTER 1

Dimensional Analysis

Topic Note: Dimensional Analysis
Dimensional analysis is a powerful tool in physics used to check the mathematical plausibility of equations, derive empirical formulas, and understand the relationship between different physical quantities. The three fundamental dimensions in mechanics are Mass (M), Length (L), and Time (T).
The Principle of Dimensional Homogeneity: You can only add, subtract, or equate quantities that have the exact same dimensions.

Reference: Common Physical Dimensions

Quantity	Symbol	Dimensional Formula	SI Unit
Mass	m	M	kg
Length	L, x, r, d	L	m
Time	t	T	s
Velocity	v	LT^{-1}	m/s
Acceleration	a, g	LT^{-2}	$\mathrm{m/s^2}$
Force	F	MLT^{-2}	N (Newton)
Energy / Work	E, W, K, U	ML^2T^{-2}	J (Joule)
Power	P	ML^2T^{-3}	W (Watt)
Pressure	P, p	$ML^{-1}T^{-2}$	Pa (Pascal)
Density	ρ	ML^{-3}	$\mathrm{kg/m^3}$
Angle	θ	1 (Dimensionless)	rad

1.1 Multiple Choice Questions

PROBLEM 1

LEVEL I

If the physical equation $X = Y + Z$ is valid, what condition must be met?

(a) The dimensions of X must equal the dimensions of Y multiplied by Z.

(b) The dimensions of X, Y, and Z must be exactly identical.

(c) Y and Z must be dimensionless.

PROBLEM 2

LEVEL I

You are trying to recall the formula for "time" t, where d is distance and v is velocity. Which of the following formulas is **impossible** to be correct?

(a) $t = \frac{d}{v}$

(b) $t = \frac{2d}{v}$

(c) $t = d \cdot v$

PROBLEM 3

LEVEL I

In the wave equation $y = \sin(k \cdot x)$, if x represents "distance", what is the dimension of the constant k?

(a) L (Length)

(b) L^{-1} (Inverse of length)

(c) Dimensionless

PROBLEM 4

LEVEL I

The area of a circle is $A = \pi \cdot r^2$. What is the dimension of the constant π?

(a) Length (L)

(b) Area (L^2)

(c) Dimensionless (1)

PROBLEM 5

LEVEL I

Hooke's Law for a spring states that the force equals the spring constant multiplied by the extension ($F = k \cdot x$). What is the standard unit of the constant k?

(a) Newton · meter

(b) Newton / meter

(c) Newton

PROBLEM 6

LEVEL I

A water tank drains at a constant rate R. What are the dimensions of this rate R?

(a) L^3T

(b) L^2T^{-1}

(c) L^3T^{-1}

PROBLEM 7

LEVEL I

Mechanical work is defined as (Force × Distance), and kinetic energy is defined as ($\frac{1}{2}\times$ Mass × Velocity squared). What is the relationship between their dimensions?

(a) The dimensions of work are greater.

(b) They have no relationship.

(c) They have exactly the same dimensions.

PROBLEM 8

LEVEL I

Which of the following physical quantities is dimensionless?

(a) Angle θ

(b) Acceleration a

(c) Density ρ

PROBLEM 9

LEVEL I

If you are told that an unknown physical quantity has the dimensions $(L \cdot T^{-2})$, what is this quantity?

(a) Velocity

(b) Acceleration

(c) Force

PROBLEM 10

LEVEL II

An object's velocity decreases according to the relation $v = v_0 \cdot \mathrm{e}^{-c \cdot t}$, where t is time and e is Euler's number. What is the dimension of the constant c?

(a) T (Time)

(b) T^{-1} (Inverse of time)

(c) L/T (Velocity)

PROBLEM 11

LEVEL III

In astrophysics, the Chandrasekhar mass limit M_{Ch} of a white dwarf depends on Planck's constant $\hbar$, the speed of light c, Newton's gravitational constant G, and the mass of a nucleon m_n. Using dimensional analysis, what is the proportional dependence of M_{Ch} on these fundamental constants?

(a) $M_{\mathrm{Ch}} \propto \left(\frac{\hbar c}{G}\right)^{3/2} m_n^{-2}$

(b) $M_{\text{Ch}} \propto \left(\frac{\hbar G}{c^3}\right)^{1/2} m_n$

(c) $M_{\text{Ch}} \propto \frac{\hbar c}{G m_n}$

(d) $M_{\text{Ch}} \propto \frac{G m_n^2}{\hbar c}$

1.2 Free-Response Questions

PROBLEM 12

LEVEL I

During a lab experiment, your lab partner proposed the following equation to calculate the velocity (v) of an object:

$$v = d + a \cdot t$$

(where d is distance, a is acceleration, and t is time).
Explain in a short paragraph, using dimensional analysis, why this equation is mathematically and physically impossible, and state the fundamental principle your partner violated.

PROBLEM 13

LEVEL II

When we use dimensional analysis to derive the kinetic energy formula, we can easily find that energy is proportional to mass and velocity squared ($m \cdot v^2$). However, this method completely fails to tell us about the fraction ($\frac{1}{2}$) in the actual formula ($E = \frac{1}{2}mv^2$).
Write a scientific explanation detailing why dimensional analysis is unable to discover pure mathematical constants and numbers.

PROBLEM 14

LEVEL II

The golden rule states: "If the dimensions of the right side do not equal the dimensions of the left side, the equation is definitely wrong." However, is the reverse true? That is, if you prove that both sides are dimensionally identical, does this necessarily mean the equation is 100% physically correct? Support your answer with a logical argument or example.

PROBLEM 15

LEVEL II

In many physical laws, we use functions like sine (sin) or logarithm (log). Explain why any quantity we place inside these functions (such as the angle θ or a variable exponent) must be a "dimensionless quantity". What would happen physically if we tried to calculate sin(5 meters) or $\log(10\text{ kg})$?

PROBLEM 16

LEVEL III

The speed v of ocean surface waves depends on the wavelength λ, fluid density ρ, gravitational acceleration g, and surface tension γ (where $[\gamma] = MT^{-2}$).

(a) Using Buckingham's Π theorem, show that the phase velocity of surface waves can be expressed as:

$$v = \sqrt{\frac{g\lambda}{2\pi} + \frac{2\pi\gamma}{\rho\lambda}}$$

(b) Identify the dominant regime for very long wavelengths ($\lambda \to \infty$) versus extremely short wavelengths ($\lambda \to 0$).

PROBLEM 17

LEVEL III

When an intense nuclear explosion occurs in air, a spherical blast wave expands outward. The radius $R(t)$ of the shock wave depends on time t, energy released E, and initial air density ρ_0.

(a) Use dimensional analysis to derive the relationship between $R(t)$ and time t.

(b) If the blast wave radius reaches $R_1 = 80$ m at time $t_1 = 1.0$ ms, estimate the radius R_2 at $t_2 = 16.0$ ms.

Part II

Kinematics II

CHAPTER 2

Kinematics II

Topic Note: Advanced Kinematics (Rolling, Relative Motion, and Constraints)
Advanced kinematics extends beyond basic trajectory equations to include reference frame transformations, rolling constraints, and circular kinematics:

- **Rolling Without Slipping:** For a cylinder or wheel of radius R rolling on a stationary surface, the velocity of the center of mass v_{cm} and angular velocity ω are related by $v_{\text{cm}} = \omega R$. The instantaneous center of rotation lies at contact points with zero local speed.
- **Relative Motion in Rotating and Accelerating Frames:** Galilean transformations for position, velocity, and acceleration allow analysis in non-inertial systems or moving constraints:
$$\vec{v}_{\text{absolute}} = \vec{v}_{\text{rel}} + \vec{V}_{\text{frame}}$$
- **Circular Kinematics:** Centripetal acceleration is given by $a_c = \frac{v^2}{r} = \omega^2 r$, and tangential acceleration by $a_t = \alpha r$.

2.1 Constraint Kinematics

PROBLEM 18

LEVEL I

Rotation Count of a Pure Rolling Drum

A uniform circular drum of radius $R = 0.50$ m rolls across a flat plane without slipping. If its central axis traverses a net distance of 31.4 m, how many full rotations did the drum complete? (Assume $\pi \approx 3.14$)

PROBLEM 19

LEVEL I

Linear Distance from Wheel Rotations

A circular wheel of diameter $D = 0.80$ m undergoes pure rolling along a horizontal road. If the

wheel turns through 450 complete revolutions, calculate the total horizontal distance traveled by the axle of the wheel.

2.2 Relative Motion and Moving Media

PROBLEM 20

LEVEL I

Relative Speed of Two Walkers

Two pedestrians walk along a straight path in the same direction. Pedestrian A walks at a steady pace of 4.0 km/h, while Pedestrian B walks ahead at 6.5 km/h. Calculate the velocity of Pedestrian B relative to Pedestrian A.

PROBLEM 21

LEVEL I

Catching Up on a Straight Highway

A highway patrol cruiser traveling west at 95.0 km/h pursues a speeder moving west at 80.0 km/h.

(a) What is the relative velocity of the speeder as observed from the patrol cruiser?

(b) What is the relative velocity of the patrol cruiser as observed from the speeder?

(c) If the vehicles are initially separated by 250 m, how long will it take for the patrol cruiser to catch the speeder?

PROBLEM 22

LEVEL I

Round-Trip Swimmer in a Current

A river flows with a uniform current of 0.500 m/s. A swimmer travels 1.00 upstream and returns to the starting location.

(a) If the swimmer maintains a speed of 1.20 m/s relative to still water, calculate the total time required for the round trip.

(b) How long would the same round trip take in perfectly still water?

(c) Explain physically why the presence of a river current always increases the total duration of a round trip.

PROBLEM 23

LEVEL II

Falling Object in an Accelerating Train

A bolt detaches from the ceiling of a train compartment that is accelerating northward along a straight track at 2.50 m/s^2.

(a) What is the acceleration vector of the bolt relative to the train?

(b) What is the acceleration vector of the bolt relative to the Earth?

(c) Describe the trajectory of the bolt as seen by a passenger seated in the train.

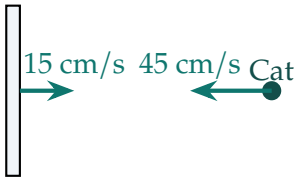
(d) Describe the trajectory of the bolt as seen by an observer standing on the ground.

PROBLEM 24

LEVEL II

Approach Rate to a Moving Plane Mirror

A cat walks directly toward a flat vertical mirror at 45 cm/s. Simultaneously, the mirror moves toward the cat at 15 cm/s. Find the rate at which the distance between the cat and its reflected image decreases.



PROBLEM 25

LEVEL II

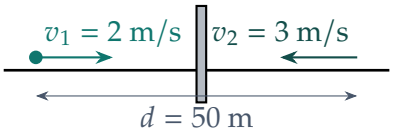
Opposing Particles with Equal Acceleration

Two particles are initially separated by distance $d = 50$ m along a straight track. Particle 1 moves right with initial speed $v_1 = 2$ m/s, while Particle 2 moves left with initial speed $v_2 = 3$ m/s. Both particles experience a constant acceleration a directed to the left. If they

collide after $t = 10$ s, find:

(a) The magnitude of acceleration a.

(b) The relative speed of the particles at the moment of impact.



PROBLEM 26

LEVEL II

Drifting Raft and Returning Motorboat

A motorboat traveling downstream along a river passes a floating log raft at point A. The boat continues downstream for $T = 60$ min, turns back, and travels upstream until it meets the raft again at distance $L = 6.0$ downstream from A. Assuming the boat maintains constant speed relative to the water, calculate the speed of the river current.

PROBLEM 27

LEVEL II

Optimal Path in a Wedge Enclosure

Two flat walls meet at an angle $2\alpha = 40°$. A runner starts at point P located along the bisector at distance $d = 10$ m from the corner. The runner must touch both walls in sequence before returning to P. If moving at $v = 2.0$ m/s, calculate the minimum time needed.

PROBLEM 28

LEVEL II

Cross-River Navigation and Minimum Drift

A river of width $W = 400$ m flows with a uniform current $v_r = 5.0$ m/s. A boat moves at speed $v_b = 3.0$ m/s relative to the water.

(a) Determine the heading angle θ relative to the bank that minimizes downstream drift.

(b) Calculate the minimum downstream drift distance experienced by the boat upon reaching the opposite bank.

PROBLEM 29

LEVEL III

Signal Timings on an Accelerating Train

A train of length $l = 420$ m accelerates from rest with $w = 2.5 \times 10^{-2}$ m/s^2. At $t_1 = 20$ s, a light at the front flashes (Event 1). At $t_2 = t_1 + \tau$ where $\tau = 45$ s, a light at the rear flashes (Event 2).

(a) Calculate the spatial distance between the two events in the train's frame.

(b) Calculate the distance between the two events in the ground frame.

(c) Find the constant velocity V of an inertial frame K where both events occur at identical spatial coordinates.

2.3 Circular Motion and Rotating Systems

PROBLEM 30

LEVEL I

Centripetal Acceleration of a Tracked Vehicle

A model vehicle travels along a circular track of radius $r = 12.0$ m at a constant speed $v = 6.0$ m/s. Calculate the magnitude of its centripetal acceleration.

PROBLEM 31

LEVEL I

Angular Speed and Period of a Rotating Ring

A point on the outer rim of a rotating ring of radius $R = 0.50$ m experiences a constant centripetal acceleration of magnitude $a_c = 18.0$ m/s^2.

(a) Compute the angular speed ω of the ring.

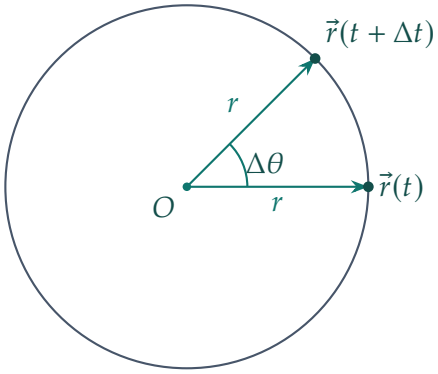
(b) Determine the period T of one complete rotation.

PROBLEM 32

LEVEL I

Geometric Derivation of Radial Acceleration

Derive the expression $a_c = v^2/r$ for uniform circular motion using vector geometry. Construct the velocity vector triangle corresponding to a small angular change $\Delta\theta$.



PROBLEM 33

LEVEL I

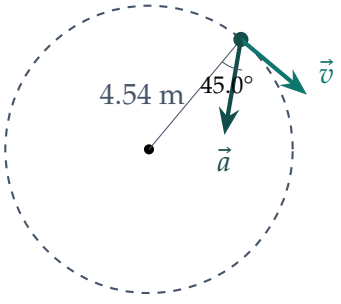
Accelerations in Non-Uniform Circular Motion

A particle moves clockwise along a circular track of radius $R = 4.54$ m. At a given instant, its total acceleration vector has magnitude $a = 5.0\ \text{m/s}^2$ and forms an angle of $45.0°$ with the inward radial direction.

(a) Calculate the radial acceleration component.

(b) Determine the instantaneous speed of the particle.

(c) Calculate the tangential acceleration component.



PROBLEM 34

LEVEL II

Accelerations on a Spinning Platform

Object A at radius $r_1 = 1.50$ m and object B at radius $r_2 = 2.50$ m rest on a turntable along the same radial line. At a given moment, object A has acceleration $\vec{a}_1 = (3.00\ \text{m/s}^2)\hat{i} + (4.00\ \text{m/s}^2)\hat{j}$. Determine the acceleration vector of object B at this moment.

PROBLEM 35

LEVEL II

Particle Slowing Down on a Curved Ramp

A particle enters a circular track of radius $R = 5.0$ m with an initial speed $v_0 = 10.0$ m/s. As it moves along the track, its tangential deceleration is constant and equal to $a_t = -2.0\ \text{m/s}^2$.

(a) Find the time t^* at which the radial and tangential accelerations have equal magnitude.

(b) Determine the distance traveled along the arc up to time t^*.

PROBLEM 36

LEVEL III

Reconstructing Circular Trajectories

A particle undergoes uniform circular motion in the xy-plane. At $t_1 = 3.00$ s, it is at $(4.00\ \text{m}, 5.00\ \text{m})$ with velocity $\vec{v}_1 = (2.50\ \text{m/s})\hat{j}$ and acceleration pointing in the $+x$ direction. At $t_2 = 8.50$ s, its velocity is $\vec{v}_2 = (-2.50\ \text{m/s})\hat{i}$ with acceleration pointing in the $+y$ direction. Assuming $t_2 - t_1$ is less than one period, find the center coordinates (x_c, y_c) of the orbit.

Part III

Dynamics I

CHAPTER 3

Dynamics I

Topic Note: Newton's Laws and Fundamental Forces
Dynamics links the motion of bodies to the forces causing it. Key principles include:

- **Newton's First Law (Law of Inertia):** A body remains at rest or continues to move in a straight line at constant velocity unless acted upon by a net external force:

$$\vec{F}_{\text{net}} = 0 \implies \vec{a} = 0 \quad (\vec{v} = \text{constant})$$

- **Newton's Second Law:** The net force acting on a body equals the rate of change of its momentum:

$$\vec{F}_{\text{net}} = \frac{\Delta \vec{p}}{\Delta t} = m\vec{a} \quad \text{(for constant mass)}$$

- **Newton's Third Law (Action-Reaction):** When one body exerts a force on a second body, the second body simultaneously exerts a force equal in magnitude and opposite in direction on the first body:

$$\vec{F}_{A\to B} = -\vec{F}_{B\to A}$$

- **Friction Forces:** Static friction opposes impending motion ($f_s \leq \mu_s N$), while kinetic friction opposes active sliding ($f_k = \mu_k N$).
- **Tension and Constraints:** Ideal strings are massless and inextensible, transmitting uniform tension forces along their length and coupling accelerations of connected bodies.

3.1 Static Equilibrium and Cable Tension

PROBLEM 37

LEVEL I

Vertical Cable Tension

A heavy light-fixture of mass $m = 12.0$ kg hangs at rest from a single vertical cord connected to the ceiling.

Find the tension T in the cord.

PROBLEM 38

LEVEL I

Horizontal Push on a Wall

A box of mass $m = 5.0$ kg is held at rest on a smooth horizontal floor. A person pushes horizontally on the box with a force $F = 20$ N towards a rigid vertical wall.

Determine the magnitude of the normal force exerted by the wall on the box.

PROBLEM 39

LEVEL I

Single Cable Weight Balance

A crate of mass $m = 25.0$ kg hangs vertically at rest from a single vertical steel cable attached to an overhead crane hook.

Determine the tension force T in the cable.

PROBLEM 40

LEVEL I

Crate of Toys on an Accelerating Sleigh

A crate of toys remains at rest on a sleigh as the sleigh is pulled up a hill with an increasing speed. The crate is not fastened down to the sleigh.

Which force is directly responsible for the crate's increase in speed up the hill?

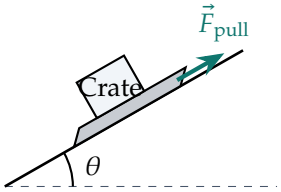
(a) The force of static friction of the sleigh on the crate

(b) The contact force (normal force) of the ground on the sleigh

(c) The contact force (normal force) of the sleigh on the crate

(d) The gravitational force acting on the sleigh

(e) No force is needed



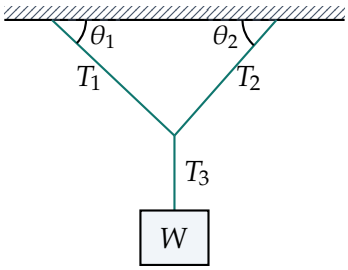
PROBLEM 41

LEVEL I

Suspended Cargo in Equilibrium

A crate of industrial supplies weighing $W = 360$ N is held in static equilibrium by three light cables joined at a central node. Two upper cables extend to a rigid overhead ceiling, forming angles $\theta_1 = 53.0°$ and $\theta_2 = 37.0°$ relative to the horizontal.

Determine the tension forces T_1, T_2, and T_3 in each supporting cable.



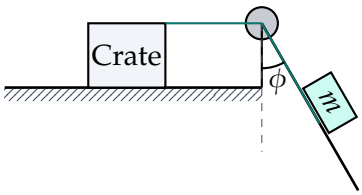
PROBLEM 42

LEVEL I

Stationary Container on Incline with Pulley

A heavy crate rests in static equilibrium on a horizontal table. A light, flexible cord attached to the crate extends horizontally over a smooth, frictionless pulley and connects to a secondary block of mass $m = 3.00$ kg that rests on an attached smooth ramp inclined at an angle $\phi = 30.0°$ relative to the vertical.

Calculate the tension force T exerted by the cord on the stationary crate.



PROBLEM 43

LEVEL II

General Cable Tension Formula

A payload of weight F_g hangs in static equilibrium suspended by three ideal wires. The two upper support wires attach to a horizontal ceiling at inclination angles θ_1 and θ_2 measured relative to the horizontal.

Prove analytically that the tension T_1 in the left wire is given by:

$$T_1 = \frac{F_g \cos\theta_2}{\sin(\theta_1 + \theta_2)}$$

PROBLEM 44

LEVEL II

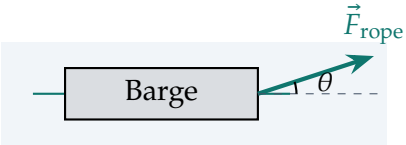
Towing a River Barge

A draft horse hauls a canal barge of mass $M = 9500$ kg along a straight waterway parallel to the positive x-axis. The towline exerts a tension force of magnitude $F = 7900$ N oriented at an angle $\theta = 18°$ relative to the positive x-axis. The barge accelerates forward along the x-axis at $a = 0.12\ \text{m/s}^2$.

Determine:

(a) The magnitude of the net water resistance force acting on the barge.

(b) The direction angle of this water force relative to the positive x-axis.



PROBLEM 45

LEVEL II

Terminal Velocity Under Quadratic Drag

A spherical raindrop of mass $m = 0.15$ g and cross-sectional area $A = 2.0 \times 10^{-4}\ \text{m}^2$ falls vertically through the atmosphere. Air resistance exerts a drag force governed by the quadratic drag formula:

$$F_d = \frac{1}{2}\rho v^2 C_d A$$

where the air density is $\rho = 1.20\ \text{kg/m}^3$ and the drag coefficient is $C_d = 0.47$.

Determine the terminal velocity v_t reached by the falling raindrop.

PROBLEM 46

LEVEL III

Charged Particle in Uniform Electric Field on an Incline

A small block of mass $m = 0.20$ kg carrying a positive charge $q = 15$ μC rests on a smooth, non-conducting plane inclined at an angle $\theta = 30.0°$ to the horizontal. A uniform electric field $\vec{E}$ of magnitude $E = 4.0 \times 10^4$ N/C is directed horizontally to the right (parallel to the base of the incline).

The electrostatic force on a charge in an electric field is given by $\vec{F}_e = q\vec{E}$.

Determine:

(a) The acceleration of the charged block up or down the incline.

(b) The magnitude of the normal force exerted by the incline on the block.

PROBLEM 47

LEVEL III

Dockside Crane Rigging Tension Analysis

Figure **??** shows the cable rigging for a crane to lift a large container with mass 2.80×10^4 kg onto or from a ship. Assume that the mass is uniformly spread within the container. The container is supported at its corners by four identical cables that are under tension T_4 and that are at an angle $\theta_4 = 60.0°$ with the vertical. They are attached to a horizontal bar that is supported by two identical cables under tension T_2 and at angle $\theta_2 = 40.0°$ with the vertical. They are attached to the main crane cable that is under tension T_1 and vertical. Assume the mass of the bar is negligible compared to the weight of the container. What are the values of (a) T_1, (b) T_2, and (c) T_4?

3.2 Multi-Body Dynamics and Coupled Systems

PROBLEM 48

LEVEL I

Single Accelerated Block

A crate of mass $m = 8.0$ kg rests on a frictionless horizontal floor. A constant horizontal force $F = 24.0$ N pulls the crate to the right.

Calculate the magnitude of the crate's acceleration.

PROBLEM 49

LEVEL I

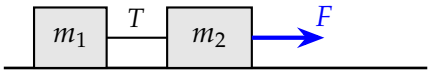
Two Blocks Connected by a String

Two blocks of masses $m_1 = 2$ kg and $m_2 = 3$ kg rest on a frictionless horizontal surface. A horizontal force $F = 15$ N is applied to block m_2, which is connected to m_1 by a light string.

Find:

(a) The acceleration of the system.

(b) The tension T in the string connecting the two blocks.



PROBLEM 50

LEVEL I

Simple Ideal Pulley System

A block of mass $m_1 = 4.0$ kg rests on a smooth horizontal table and is attached to a light cord that passes over a frictionless pulley. The other end of the cord supports a vertically hanging mass $m_2 = 2.0$ kg.

Determine the magnitude of the acceleration of the system when released from rest.

PROBLEM 51

LEVEL I

Two Blocks Pushed on a Frictionless Surface

Two blocks with masses $m_1 = 3.0$ kg and $m_2 = 2.0$ kg are placed in contact on a horizontal frictionless surface. A horizontal force $F = 15.0$ N pushes block m_1 directly.

Find:

(a) The acceleration of the two blocks.

(b) The contact force between block m_1 and block m_2.

PROBLEM 52

LEVEL I

Connected Blocks on a Frictionless Surface

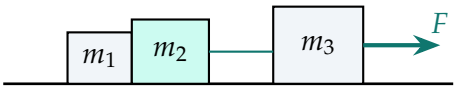
Three blocks with masses $m_1 = 1.5$ kg, $m_2 = 2.5$ kg, and $m_3 = 4.0$ kg are placed on a smooth, horizontal track. Blocks m_1 and m_2 are in contact, and m_2 is connected to m_3 by a light cord. A constant horizontal force $F = 48.0$ N pulls the lead block m_3 forward.

Calculate:

(a) The common acceleration of the combined three-block assembly.

(b) The tension in the coupling cord between m_3 and m_2.

(c) The contact force transmitted between m_2 and m_1.



PROBLEM 53

LEVEL II

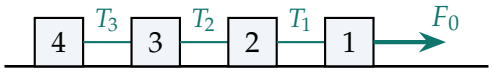
Tension in a Train of 4 Connected Blocks

A collection of 4 identical blocks, each of mass m, are connected by unstretchable ropes of negligible mass on a horizontal surface. An external force F_0 acts on Block 1, pulling horizontally to the right.

Find the tension T_1, T_2, and T_3 in the ropes connecting the blocks for:

(a) The case where there is no friction between the blocks and the surface.

(b) The case where the coefficient of kinetic friction between each block and the surface is μ_k (assuming F_0 is large enough to accelerate the blocks).



PROBLEM 54

LEVEL II

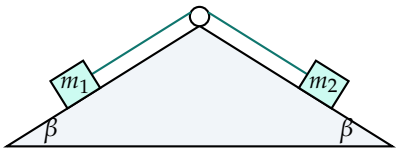
Symmetrical Double-Incline System

Two blocks of masses $m_1 = 4.00$ kg and $m_2 = 7.50$ kg are connected by a light string passing over a frictionless apex pulley atop a double inclined wedge. Both inclined faces form an angle $\beta = 32.0°$ with the horizontal and are frictionless.

Determine:

(a) The magnitude and direction of the acceleration of each block.

(b) The tension force in the connecting string.



PROBLEM 55

LEVEL II

Electrostatic Repulsion of Suspended Charges

Two identical small spheres, each of mass $m = 5.0$ g, are suspended from a common point by two light insulating strings of length $L = 0.50$ m. When given identical positive charges q, the spheres repel each other and hang in equilibrium separated by a horizontal distance $d = 6.0$ cm.

Using Coulomb's law for electrostatic force, $F_e = k_e \frac{q^2}{r^2}$, determine the magnitude of the charge q on each sphere. (Take $k_e = 8.99 \times 10^9$ N m^2/C^2 and $g = 9.81$ m/s^2).

PROBLEM 56

LEVEL II

Identical Mass Atwood System with Table Friction

Three identical blocks, each of mass m, are connected as shown: block m_0 hangs vertically from an ideal string passing over a frictionless pulley at the edge of a horizontal surface, while blocks m_1 and m_2 lie in tandem on the horizontal surface, linked by a second string. The coefficient of friction between the horizontal surface and blocks m_1 and m_2 is k.

(a) Determine the downward acceleration a of mass m_0.

(b) Find the tension T_{12} in the connecting cord between masses m_1 and m_2.

(c) Specify the threshold condition on k for which motion occurs.

PROBLEM 57

LEVEL II

Contact Force Between Adjacent Blocks on an Incline

Two rectangular blocks 1 and 2 with masses m_1 and m_2 are placed in direct contact on an inclined plane forming an angle α with the horizontal. The coefficients of friction between the incline and blocks 1 and 2 are k_1 and k_2, respectively.

(a) Find the condition under which the blocks push against each other during motion down the incline.

(b) Determine the acceleration of the combined blocks and the normal force of contact between them when this condition is satisfied.

PROBLEM 58

LEVEL II

Atwood Machine in an Accelerating Elevator

A frictionless, light pulley is attached to the ceiling of an elevator cabin. An ideal string passing over the pulley supports two masses, m_1 and m_2. The elevator starts ascending with a constant upward acceleration a_0.

(a) Calculate the acceleration of mass m_1 relative to the elevator car and relative to the fixed shaft frame.

(b) Determine the total vertical force exerted by the pulley support on the elevator ceiling.

PROBLEM 59

LEVEL III

Accelerated Double-Pulley Atwood System

A light, frictionless pulley A is suspended from a rigid ceiling. A light, inextensible string passes over pulley A, carrying a block of mass m_1 at one end and a second light, frictionless pulley B at the other end.

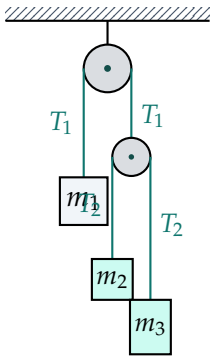
A second light string passes over pulley B and supports two blocks with masses m_2 and m_3 ($m_3 > m_2$). The entire system is released from rest in a uniform gravitational field g.

(a) Determine the kinematic constraint equations relating the accelerations a_1, a_2, and a_3 of the three masses with respect to the ground frame.

(b) Derive the expression for the acceleration a_1 of mass m_1 in terms of m_1, m_2, m_3, and g.

(c) Find the tension forces T_1 and T_2 in both strings.

(d) State the condition on the masses m_1, m_2, and m_3 for which pulley B remains stationary in space upon release.



PROBLEM 60

LEVEL III

Table-Edge Pulley System with Nested Second Pulley

An object of mass m_1 hangs from a string that passes over a very light fixed pulley P_1. The string connects to a second very light pulley P_2. A second string passes around this pulley with one end attached to a wall and the other to an object of mass m_2 on a frictionless, horizontal table. (a) If a_1 and a_2 are the accelerations of m_1 and m_2, respectively, what is the relation between these accelerations? Find expressions for (b) the tensions in the strings and (c) the accelerations a_1 and a_2 in terms of the masses m_1 and m_2, and g.

PROBLEM 61

LEVEL III

Accelerated Block Assembly on a Moving Block

A large block of mass M rests on a frictionless horizontal plane. Mounted on top of M is a frictionless horizontal ledge terminating in a pulley, which suspends a hanging mass m_1 vertically along the right side of M. A second mass m_2 rests on top of M and connects to m_1 via a light cord running over the pulley.

Determine the horizontal driving force F that must be applied to M so that masses m_1 and m_2 remain completely stationary relative to M.

PROBLEM 62

LEVEL III

Modified Double-Pulley System on an Incline

A block of mass m_1 slides on a smooth plane inclined at angle α. A cord attached to m_1 runs

parallel to the incline, passes over a fixed pulley at the top, and connects to a second block of mass $m_2 = \eta m_1$ hanging vertically.

Determine the acceleration of block m_2 for all possible cases depending on the mass ratio η and inclination α.

PROBLEM 63

LEVEL III

Triple Mass System with Compound Pulley

In a frictionless mechanism, mass m_0 rests on a smooth horizontal table, connected via strings and pulleys to two suspended masses m_1 and m_2.

Find the acceleration of mass m_1 in terms of m_0, m_1, m_2, and g.

3.3 Friction, Sliding, and Inclines

PROBLEM 64

LEVEL I

Normal Force on an Incline

A block of mass $m = 6.0$ kg rests on a smooth incline angled at $\theta = 30.0°$ relative to the horizontal.

Calculate the magnitude of the normal force N exerted by the incline on the block.

PROBLEM 65

LEVEL I

Kinetic Friction on a Flat Surface

A box of mass $m = 10.0$ kg slides across a horizontal rough floor with an initial speed $v_0 = 5.0$ m/s. The coefficient of kinetic friction between the box and the floor is $\mu_k = 0.25$.

Find the magnitude of the deceleration caused by friction.

PROBLEM 66

LEVEL I

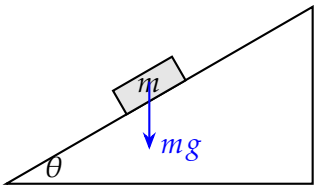
Block on an Inclined Plane

A block of mass $m = 4$ kg rests on a frictionless plane inclined at an angle $\theta = 30°$ to the horizontal.

Find:

(a) The component of the gravitational force acting down the incline.

(b) The acceleration of the block as it slides down the plane.



PROBLEM 67

LEVEL II

Threshold Friction under Normal Loading

A block of mass $m = 3.00$ kg rests on a horizontal table. A constant horizontal force $F = 8.00$ N is applied to it. In addition, a vertical downward force P is exerted on top of the block. The coefficients of friction between block and table are $\mu_s = 0.45$ and $\mu_k = 0.30$.

Determine the magnitude of the resulting friction force acting on the block when P is set to:

(a) $P = 5.00$ N

(b) $P = 12.0$ N

(c) $P = 20.0$ N

PROBLEM 68

LEVEL II

Cargo Cart and Incline Launch

A freight cart with an open top and mass $M = 60.0$ kg rolls freely on a level frictionless track to the left at speed $v_0 = 4.50$ m/s. A parcel of mass $m = 12.0$ kg slides down a chute angled at $\theta = 30.0°$ to the horizontal, exiting the chute's lower edge with a speed $u_0 = 4.00$ m/s. The exit point of the chute is located at height $h = 3.50$ m above the bed of the cart.

(a) Determine the speed of the parcel immediately before landing in the cart.

(b) Compute the final horizontal speed of the combined cart and parcel after collision.

PROBLEM 69

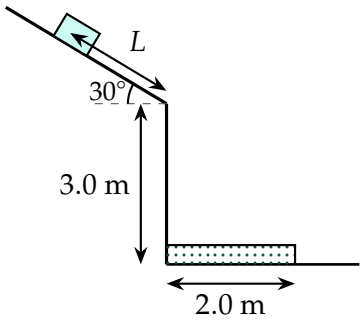
LEVEL II

Box Sliding off a Roof into a Flower Bed

A box of nails begins sliding from rest on the roof of a house. The roof makes an angle of 30° above the horizontal. The coefficient of sliding friction is $\mu_k = 1/4$. The box slides off the edge of the roof with a speed of 3.5 m/s.

(a) What is the distance L the box slides on the roof before falling off? (Use $g = 9.8\ \text{m/s}^2$).

(b) The flower bed extends from the side of the house and is 2.0 m wide. The edge of the roof is 3.0 m above the ground. Does the box of nails land in the flowers? Justify your answer with a calculation that shows where the box lands.



PROBLEM 70

LEVEL II

Non-Hookian Spring on an Incline

A non-Hookian spring exerts a restoring force $F = -kx^2$, where k is a positive constant and x is the displacement from its unstretched position. A mass m connected to this spring (initially unstretched and at rest) is released on a frictionless ramp inclined at 30° relative to the horizontal, with the cord passing over an ideal pulley.

How far x does the spring extend before the system momentarily comes to rest?

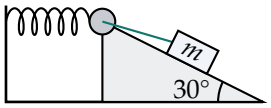
(a) $\left(\frac{3mg}{2k}\right)^{1/2}$

(b) $\left(\frac{mg}{k}\right)^{1/2}$

(c) $\left(\frac{2mg}{k}\right)^{1/2}$

(d) $\left(\frac{\sqrt{3}mg}{k}\right)^{1/3}$

(e) $\left(\frac{3\sqrt{3}mg}{2k}\right)^{1/3}$



PROBLEM 71

LEVEL II

Tabletop Rotating Puck with Hanging Cylinder

A puck of mass $m = 1.50$ kg slides in a circle of radius $r = 20.0$ cm on a frictionless table while attached to a hanging cylinder of mass $M = 2.50$ kg by means of a cord that extends through a hole in the table (Figure **??**). What speed keeps the cylinder at rest?

PROBLEM 72

LEVEL II

Rock Climber in a Chimney

A 49 kg rock climber is climbing a "chimney." The coefficient of static friction between her shoes and the rock is 1.2; between her back and the rock is 0.80. She has reduced her push against the rock until her back and her shoes are on the verge of slipping. (a) Draw a free-body diagram of her. (b) What is the magnitude of her push against the rock? (c) What fraction of her weight is supported by the frictional force on her shoes?

PROBLEM 73

LEVEL III

Accident Investigation on an Inclined Roadway

A forensic accident investigator analyzes a rear-end collision on a downhill road sloping at an angle $\theta = 14.0°$ below the horizontal. Car A locked its brakes and skidded a distance $d = 28.0$ m down the slope before striking stationary Car B. Car A's initial velocity when braking began was $v_0 = 20.0$ m/s.

Calculate the impact velocity of Car A for the following road conditions:

(a) Dry asphalt with kinetic friction coefficient $\mu_k = 0.65$.

(b) Wet, leaf-covered asphalt with kinetic friction coefficient $\mu_k = 0.12$.

PROBLEM 74

LEVEL III

Crate Pressed Against a Truck

A crate of mass m rests against the flat vertical front face of a heavy delivery truck. The coefficient of static friction between the crate and the truck's front surface is μ.

Find the minimum horizontal forward acceleration a_{min} of the truck necessary to prevent the crate from slipping downward under gravity.

PROBLEM 75

LEVEL III

Stacked Block and Slab System

A heavy slab of mass $M = 12.0\,\text{kg}$ lies on a smooth horizontal floor. A block of mass $m = 8.00\,\text{kg}$ rests on top of the slab. A horizontal force $F = 110\,\text{N}$ is applied directly to the bottom slab. The coefficient of static friction between block and slab is $\mu_s = 0.40$, and kinetic friction is $\mu_k = 0.25$.

Analyze the motion: determine whether the block slides relative to the slab, and calculate the resulting accelerations a_{slab} and a_{block}.

PROBLEM 76

LEVEL III

Tabletop Incline Projectile Launch

A small block of mass $m = 1.50\,\text{kg}$ is released from rest at a vertical height $h = 0.600\,\text{m}$ above a tabletop, sliding down a smooth ramp inclined at $\theta = 30.0^\circ$. The bottom of the ramp sits at the edge of the tabletop, which is elevated at height $H = 1.80\,\text{m}$ above the floor.

(a) Determine the acceleration of the block along the ramp.

(b) Calculate the velocity vector of the block as it leaves the ramp edge.

(c) Find the horizontal range R from the table edge to where the block impacts the floor.

(d) Compute the total time elapsed from release until impact with the floor.

(e) State whether varying the mass m would alter any of these physical quantities.

PROBLEM 77

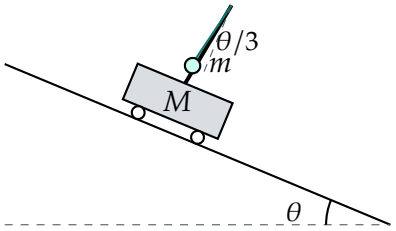
LEVEL III

Accelerometer on an Inclined Cart

An accelerometer consists of a ball of mass m attached to one end of a string. The other end of the string is attached to a cart of mass M. When the cart rolls down an inclined plane of angle θ with respect to the horizontal, the string makes a constant angle of $\theta/3$ with respect to the normal to the inclined plane. Express your answers in terms of known and given quantities (g, θ). Assume air resistance is negligible.

(a) What is the acceleration of the cart down the incline?

(b) What is the coefficient of kinetic friction μ between the cart and the plane?



PROBLEM 78

LEVEL III

Friction-Coupled Rod and Sliding Ball

A smooth, light pulley supports a flexible cord. One end of the cord is attached to a vertical rod of mass M, and the other end carries a small heavy ball of mass m ($M > m$). The ball has a central bore allowing it to slide along the cord with a constant friction force f. Initially, the ball is positioned directly opposite the lower tip of the rod. Upon release from rest, both bodies begin moving with constant accelerations.

Find the magnitude of the friction force f if the ball reaches the upper tip of the rod of length L after a time interval Δt.

PROBLEM 79

LEVEL III

Time to Traverse Hanging Rod

A spherical mass m_1 is attached to a string passing over fixed pulleys and positioned level with

the lower end of a vertical hollow pipe of mass $m_2 = \frac{1}{2}m_1$ and length $L = 1.20$ m. The system is released from rest.

Calculate the time required for mass m_1 to reach the level of the upper end of the pipe.

PROBLEM 80

LEVEL III

Maximum Ascent of Pulley System Mass

In a dual-pulley setup, mass m_1 is $n = 3.5$ times heavier than mass m_2. Mass m_2 is initially situated at height $h = 25.0$ cm above a stopping surface. The system is released from rest.

Compute the maximum vertical height above its initial position reached by body m_2.

CHAPTER 4

Hints

The hints below correspond to all problems across the three parts of this collection. They are provided separately so that students can attempt the problems independently before consulting guidance.

Hint 1 | Problem 1 – Problem 1

Answer: (b)
The Idea: You cannot add "velocity" to "mass" in physics. Addition and subtraction require quantities to be of the exact same type.

Hint 2 | Problem 2 – Problem 2

Answer: (c)
The Idea: Dimensional analysis catches wrong formulas instantly. Multiplying distance (L) by velocity (L/T) gives L^2/T, which is not time.

Hint 3 | Problem 3 – Problem 3

Answer: (b)
The Idea: Anything inside a trigonometric function must be a pure, dimensionless angle. To cancel out the length dimension of x, k must be the inverse of length.

Hint 4 | Problem 4 – Problem 4

Answer: (c)
The Idea: Pure numbers (π, fractions, mathematical constants) are "ghosts" in the realm of dimensions; they have no physical dimensions and do not affect the dimensional balance.

Hint 5 | Problem 5 – Problem 5

Answer: (b)
The Idea: You can rearrange any equation to find the unit of an unknown. $k = F/x$, so its unit is simply the unit of force divided by the unit of distance.

Hint 6 | Problem 6 – Problem 6

Answer: (c)
The Idea: The word "rate" in physics almost always implies "divided by time". The volume of water is length cubed (L^3), so its rate of change is volume divided by time.

Hint 7 | Problem 7 – Problem 7

Answer: (c)
The Idea: All forms of energy in the universe share the exact same dimensional footprint (ML^2T^{-2}). Dimensions unify different physical concepts.

Hint 8 | Problem 8 – Problem 8

Answer: (a)
The Idea: An angle is mathematically defined as arc length divided by radius (Length $\div$ Length). The dimensions cancel out, leaving a pure dimensionless ratio.

Hint 9 | Problem 9 – Problem 9

Answer: (b)
The Idea: Dimensions are the DNA of physical quantities. Length divided by time squared (m/s^2) is the specific footprint of acceleration.

Hint 10 | Problem 10 – Problem 10

Answer: (b)
The Idea: Exponents must be dimensionless. To cancel out the dimension of time t, c must be the inverse of time.

Hint 11 | Problem 11 – Problem 11

Answer: (a)
The Idea: Set $M_{\text{Ch}} \propto \hbar^a c^b G^c m_n^d$. The dimensions are: $[\hbar] = ML^2T^{-1}$, $[c] = LT^{-1}$, $[G] = M^{-1}L^3T^{-2}$, $[m_n] = M$. Constructing mass $[M^1L^0T^0]$: $M : a - c + d = 1$
$L : 2a + b + 3c = 0$
$T : -a - b - 2c = 0$
Adding L and T equations yields $a + c = 0 \implies c = -a$. Substituting $c = -a$ into L gives $b = a$. Then M gives $a - (-a) + d = 1 \implies 2a + d = 1$. For $a = 3/2$, we get $c = -3/2$, $b = 3/2$, and $d = -2$. Thus $M_{\text{Ch}} \propto \hbar^{3/2} c^{3/2} G^{-3/2} m_n^{-2} = \left(\frac{\hbar c}{G}\right)^{3/2} m_n^{-2}$.

Hint 12 | Problem 12 – Problem 12

Check the dimensions of each term. $[v] = LT^{-1}$. The first term on the right is $[d] = L$. The second is $[a \cdot t] = (LT^{-2})(T) = LT^{-1}$. You cannot mathematically add a length (L) to a velocity (LT^{-1}). This violates the Principle of Dimensional Homogeneity.

Hint 13 | Problem 13 – Problem 13

Dimensional analysis only tracks the fundamental physical nature of quantities (Mass, Length, Time). Pure numbers and fractions are dimensionless; they act as mathematical constants that scale the equation but do not alter its physical units. Therefore, they are "invisible" to dimensional balancing.

Hint 14 | Problem 14 – Problem 14

No, the reverse is not true. A dimensionally correct equation might still be physically wrong because it could be missing a dimensionless constant (like 2π or $1/2$), or it might include a term that has correct units but no physical relevance to the phenomenon.

Hint 15 | Problem 15 – Problem 15

Transcendental functions are defined by infinite power series (e.g., Maclaurin series). If x had a dimension (like meters), you would be adding meters to cubic meters, which violates dimensional homogeneity.

Hint 16 | Problem 16 – Problem 16

Mathematical Solution:

Part (a): The physical variables involved are $v, \lambda, \rho, g, \gamma$ ($N = 5$ variables, $K = 3$ core dimensions M, L, T). By Buckingham's Π theorem, there are $N - K = 2$ dimensionless groups (Π_1, Π_2). Selecting repeated variables ρ, g, λ:

$$\Pi_1 = \frac{v}{\sqrt{g\lambda}} \implies v \propto \sqrt{g\lambda}$$

$$\Pi_2 = \frac{\gamma}{\rho g \lambda^2} \implies \text{Bond number variation}$$

Combining the two distinct restoring forces (gravity g and surface tension $\gamma/\rho\lambda$) gives the total wave velocity squared:

$$v^2 = C_1 g\lambda + C_2 \frac{\gamma}{\rho\lambda}$$

Exact hydrodynamic analysis proves $C_1 = \frac{1}{2\pi}$ and $C_2 = \frac{2\pi}{1}$.

Part (b):

- For $\lambda \to \infty$ (gravity waves): $v \approx \sqrt{\frac{g\lambda}{2\pi}}$ (gravity dominates).
- For $\lambda \to 0$ (capillary waves): $v \approx \sqrt{\frac{2\pi\gamma}{\rho\lambda}}$ (surface tension dominates).

Hint 17 | Problem 17 – Problem 17

Mathematical Solution:

Part (a): Assume $R \propto E^a \rho_0^b t^c$. Dimensions: $[R] = L$, $[E] = ML^2T^{-2}$, $[\rho_0] = ML^{-3}$, $[t] = T$. Matching dimensions $[M^0L^1T^0]$:

$$M : a + b = 0 \implies b = -a$$

$$L : 2a - 3b = 1 \implies 2a - 3(-a) = 1 \implies 5a = 1 \implies a = 1/5, b = -1/5$$

$$T : -2a + c = 0 \implies c = 2a = 2/5$$

Therefore:

$$R(t) = C \cdot \left(\frac{E}{\rho_0}\right)^{1/5} t^{2/5}$$

Part (b): Since $R(t) \propto t^{2/5}$:

$$\frac{R_2}{R_1} = \left(\frac{t_2}{t_1}\right)^{2/5} = (16)^{2/5} = (2^4)^{2/5} = 2^{8/5} \approx 3.031$$

$$R_2 = 80 \text{ m} \times 3.031 \approx 242.5 \text{ m}$$

Hint 18 | Problem 18 – Rotation Count of a Pure Rolling Drum

Relate the linear displacement of the center of mass to the angular displacement: $\Delta x = \theta R$, where $\theta = 2\pi N$ and N is the total number of revolutions.

Hint 19 | Problem 19 – Linear Distance from Wheel Rotations

The distance covered during pure rolling is equal to the total arc length swept along the perimeter: $x = N(\pi D)$.

Hint 20 | Problem 20 – Relative Speed of Two Walkers

Use relative velocity in one dimension: $v_{B/A} = v_B - v_A$.

Hint 21 | Problem 21 – Catching Up on a Straight Highway

Define a coordinate system with west as positive. Express relative velocities as vector differences and compute time using $t = d/v_{\text{rel}}$.

Hint 22 | Problem 22 – Round-Trip Swimmer in a Current

The effective speed upstream is $v_{\text{rel}} - v_{\text{river}}$ and downstream is $v_{\text{rel}} + v_{\text{river}}$. Calculate $t = \frac{d}{v_{\text{up}}} + \frac{d}{v_{\text{down}}}$.

Hint 23 | Problem 23 – Falling Object in an Accelerating Train

In an accelerating frame, include pseudo-accelerations. Relative to Earth, only gravity acts on the free-falling bolt once detached.

Hint 24 | Problem 24 – Approach Rate to a Moving Plane Mirror

Work in the mirror's reference frame to determine the cat's relative velocity, then double this result to find the image closing speed.

Hint 25 | Problem 25 – Opposing Particles with Equal Acceleration

Since both particles experience the exact same acceleration vector, their relative acceleration is zero.

Hint 26 | Problem 26 – Drifting Raft and Returning Motorboat

Switch to a reference frame attached to the moving water. The raft remains at rest, and the boat takes equal time to move away and return.

Hint 27 | Problem 27 – Optimal Path in a Wedge Enclosure

Construct mirror reflections of point P across both walls to convert the triangular path into a straight line.

Hint 28 | Problem 28 – Cross-River Navigation and Minimum Drift

Since $v_b < v_r$, straight cross-river travel is impossible. Maximize the angle of the net velocity vector relative to the shore using geometric vector construction.

Hint 29 | Problem 29 – Signal Timings on an Accelerating Train

Compute ground positions for the front at t_1 and the rear at t_2, then apply Galilean transformations.

Hint 30 | Problem 30 – Centripetal Acceleration of a Tracked Vehicle

Apply the centripetal acceleration formula: $a_c = v^2/r$.

Hint 31 | Problem 31 – Angular Speed and Period of a Rotating Ring

Use $a_c = \omega^2 R$ to determine ω, then find period using $T = 2\pi/\omega$.

Hint 32 | Problem 32 – Geometric Derivation of Radial Acceleration

Show that the position triangle and velocity triangle for time interval Δt are similar isosceles triangles.

Hint 33 | Problem 33 – Accelerations in Non-Uniform Circular Motion

Decompose acceleration into $a_r = a\cos\theta$ and $a_t = a\sin\theta$, then relate radial acceleration to speed via $a_r = v^2/R$.

Hint 34 | Problem 34 – Accelerations on a Spinning Platform

Since both objects share the platform's rigid rotation rate ω, centripetal acceleration scales linearly with radius.

Hint 35 | Problem 35 – Particle Slowing Down on a Curved Ramp

Express speed as a function of time $v(t) = v_0 - |a_t|t$. Set radial acceleration $a_r = v(t)^2/R$ equal to $|a_t|$.

Hint 36 | Problem 36 – Reconstructing Circular Trajectories

Acceleration vectors point directly toward the center. Use velocity vectors to verify the turning sense of the path.

Hint 37 | Problem 37 – Vertical Cable Tension

Apply vertical equilibrium $\sum F_y = 0$: $T - mg = 0$.

Hint 38 | Problem 38 – Horizontal Push on a Wall

In horizontal equilibrium ($\sum F_x = 0$), the normal force from the wall balances the applied push force: $N = F$.

Hint 39 | Problem 39 – Single Cable Weight Balance

Apply vertical static equilibrium $\sum F_y = 0$, balancing tension T against the gravitational weight $W = mg$.

Hint 40 | Problem 40 – Crate of Toys on an Accelerating Sleigh

Consider the forces acting parallel to the incline specifically on the crate. The only contact force parallel to the sleigh's surface preventing relative sliding and providing upward acceleration is static friction.

Hint 41 | Problem 41 – Suspended Cargo in Equilibrium

Set up equilibrium conditions for the central node ($\sum F_x = 0$ and $\sum F_y = 0$). Solve the system of linear equations for T_1 and T_2.

Hint 42 | Problem 42 – Stationary Container on Incline with Pulley

Examine the equilibrium of the block on the smooth incline. The tension along the ideal string must balance the component of the block's weight parallel to the incline: $T = mg \sin\theta = mg \cos\phi$.

Hint 43 | Problem 43 – General Cable Tension Formula

Express T_2 in terms of T_1 using horizontal equilibrium ($T_1 \cos\theta_1 = T_2 \cos\theta_2$). Substitute into vertical equilibrium ($T_1 \sin\theta_1 + T_2 \sin\theta_2 = F_g$) and apply the trigonometric identity $\sin(\theta_1 + \theta_2) = \sin\theta_1 \cos\theta_2 + \cos\theta_1 \sin\theta_2$.

Hint 44 | Problem 44 – Towing a River Barge

Apply Newton's second law along x ($F\cos\theta + F_{\text{water},x} = Ma$) and y ($F\sin\theta + F_{\text{water},y} = 0$). Find the resultant magnitude and direction using vector components.

Hint 45 | Problem 45 – Terminal Velocity Under Quadratic Drag

At terminal velocity, acceleration is zero, so the gravitational force balances the drag force: $mg = \frac{1}{2}\rho v_t^2 C_d A$. Solve for v_t.

Hint 46 | Problem 46 – Charged Particle in Uniform Electric Field on an Incline

Resolve the horizontal electric force into components parallel and perpendicular to the incline. Parallel to the incline, the net force is $F_{\parallel} = qE\cos\theta - mg\sin\theta = ma$. Perpendicularly, $N = mg\cos\theta + qE\sin\theta$.

Hint 47 | Problem 47 – Dockside Crane Rigging Tension Analysis

Use vertical equilibrium at each connection point. For the container, $4T_4\cos\theta_4 = Mg$. For the horizontal bar, $2T_2\cos\theta_2 = 4T_4\cos\theta_4 = Mg$. For the main hook/cable connection, $T_1 = 2T_2\cos\theta_2 = Mg$.

Hint 48 | Problem 48 – Single Accelerated Block

Apply Newton's second law along the horizontal axis: $F = ma$.

Hint 49 | Problem 49 – Two Blocks Connected by a String

Treat both blocks together to find the system acceleration first, then isolate one block to find the tension.

Hint 50 | Problem 50 – Simple Ideal Pulley System

Consider both blocks as a single system of mass $(m_1 + m_2)$ driven by the weight of the hanging block $m_2 g$.

Hint 51 | Problem 51 – Two Blocks Pushed on a Frictionless Surface

Calculate acceleration using total mass $a = \frac{F}{m_1+m_2}$. Then apply Newton's second law to m_2 alone to find contact force $N_c = m_2 a$.

Hint 52 | Problem 52 – Connected Blocks on a Frictionless Surface

Treat the three blocks as a single system of total mass $M_{\text{tot}} = m_1 + m_2 + m_3$ to find acceleration $a = F/M_{\text{tot}}$. Then isolate individual blocks to determine internal contact/tension forces.

Hint 53 | Problem 53 – Tension in a Train of 4 Connected Blocks

The total mass of the system is $4m$. Analyze the overall acceleration of the 4-block system first. Then, isolate trailing blocks to solve for each rope's tension (T_3 supports Block 4, T_2 supports Blocks 3 and 4, and T_1 supports Blocks 2, 3, and 4).

Hint 54 | Problem 54 – Symmetrical Double-Incline System

Write the equations of motion along each incline: $m_2 g \sin\beta - T = m_2 a$ and $T - m_1 g \sin\beta = m_1 a$. Sum them to isolate a.

Hint 55 | Problem 55 – Electrostatic Repulsion of Suspended Charges

Analyze the equilibrium of one sphere. The forces acting on it are tension T, gravity mg, and the electrostatic repulsion $F_e = \frac{1}{4\pi\varepsilon_0}\frac{q^2}{d^2}$. Use geometry $\tan\theta = \frac{d/2}{\sqrt{L^2-(d/2)^2}}$ to equate force components: $F_e = mg\tan\theta$.

Hint 56 | Problem 56 – Identical Mass Atwood System with Table Friction

Write Newton's second law for the system of three equal masses m: $mg - 2kmg = 3ma$. For internal tension T_{12}, isolate block m_2: $T_{12} - kmg = ma$.

Hint 57 | Problem 57 – Contact Force Between Adjacent Blocks on an Incline

Block 2 tends to push block 1 if its unconstrained acceleration exceeds that of block 1, which requires $k_2 < k_1$. Set up coupled equations for m_1 and m_2 including the contact force N_c.

Hint 58 | Problem 58 – Atwood Machine in an Accelerating Elevator

Work in the non-inertial frame of the elevator, where an effective gravitational field $g_{\text{eff}} = g + a_0$ acts downward. The relative acceleration is $a_{\text{rel}} = \frac{m_1 - m_2}{m_1 + m_2} g_{\text{eff}}$.

Hint 59 | Problem 59 – Accelerated Double-Pulley Atwood System

Express the acceleration of m_2 and m_3 relative to pulley B as a_{rel}. Notice that pulley B is massless, so $T_1 = 2T_2$. Replace the system attached to pulley B with an effective reduced mass $m_{\text{eff}} = \frac{4m_2 m_3}{m_2 + m_3}$.

Hint 60 | Problem 60 – Table-Edge Pulley System with Nested Second Pulley

Analyze the kinematic constraint: as mass m_2 moves a distance x_2 on the table, pulley P_2 moves by x_2, and the string wrapped around it means $x_1 = 2x_2$, so $a_1 = 2a_2$. Set up equations of motion for m_1 ($m_1 g - T_1 = m_1 a_1$) and m_2 ($T_2 = m_2 a_2$, where $T_1 = 2T_2$ due to the movable pulley P_2).

Hint 61 | Problem 61 – Accelerated Block Assembly on a Moving Block

For m_1 not to slide vertically relative to M, the string tension must balance gravity: $T = m_1 g$. For m_2 not to slide horizontally relative to M, it must accelerate horizontally with M at acceleration $a = T/m_2 = \frac{m_1 g}{m_2}$. Apply $F = (M + m_1 + m_2)a$.

Hint 62 | Problem 62 – Modified Double-Pulley System on an Incline

Set up the net force along the direction of motion: $(m_2 - m_1 \sin\alpha)g = (m_1 + m_2)a$. Check the sign to determine if m_2 moves downward or upward.

Hint 63 | Problem 63 – Triple Mass System with Compound Pulley

Express the kinematic constraint relating the displacements of the three masses. Write Newton's second law for each body and solve the system of linear equations simultaneously.

Hint 64 | Problem 64 – Normal Force on an Incline

Resolve gravity into components. Perpendicular to the incline, equilibrium gives $N = mg\cos\theta$.

Hint 65 | Problem 65 – Kinetic Friction on a Flat Surface

The normal force is $N = mg$, making the kinetic friction $f_k = \mu_k mg$. Apply $f_k = ma$ to find acceleration.

Hint 66 | Problem 66 – Block on an Inclined Plane

Resolve the weight vector mg into components parallel and perpendicular to the inclined surface. Only the parallel component causes acceleration.

Hint 67 | Problem 67 – Threshold Friction under Normal Loading

Calculate the normal force $N = mg + P$ and the maximum static friction $f_{s,\max} = \mu_s N$. If $F \le f_{s,\max}$, the block remains static and $f = F$. If $F > f_{s,\max}$, kinetic friction applies: $f = f_k = \mu_k N$.

Hint 68 | Problem 68 – Cargo Cart and Incline Launch

Use conservation of energy to find the speed of the parcel just before landing: $v_p = \sqrt{u_0^2 + 2gh}$. For the final cart speed, apply horizontal momentum conservation: $Mv_0 - mu_0\cos\theta = (M + m)V_f$.

Hint 69 | Problem 69 – Box Sliding off a Roof into a Flower Bed

For part (a), the acceleration along the roof is $a = g(\sin 30^\circ - \mu_k \cos 30^\circ)$. Use $v^2 = 2aL$. For part (b), calculate projectile flight time t using vertical displacement $y(t) = -3.0\ \text{m}$ with initial velocity components $v_{0x} = 3.5\cos 30^\circ$ and $v_{0y} = -3.5\sin 30^\circ$, then find horizontal distance $x = v_{0x}t$.

Hint 70 | Problem 70 – Non-Hookian Spring on an Incline

Apply the Work-Energy Theorem $\Delta K = W_{\text{net}} = 0$. The potential energy stored in a spring with force $F = -kx^2$ is $U_s = \int_0^x kx'^2 \mathrm{d}x' = \frac{1}{3}kx^3$. Work done by gravity down the incline is

$W_g = mgx\sin 30^\circ = \frac{1}{2}mgx$. Equate $W_g = U_s$.

Hint 71 | Problem 71 – Tabletop Rotating Puck with Hanging Cylinder

For the hanging cylinder M to remain at rest, the string tension must balance its weight: $T = Mg$. For the puck m, this same tension provides the centripetal force: $T = \frac{mv^2}{r}$. Equate $Mg = \frac{mv^2}{r}$ and solve for v.

Hint 72 | Problem 72 – Rock Climber in a Chimney

Let F_p be her horizontal push against each wall. The normal forces from both the front and back walls equal F_p. Static friction forces at the feet and back are $f_{s,\text{shoes}} = \mu_{\text{shoes}}N_{\text{shoes}} = 1.2F_p$ and $f_{s,\text{back}} = \mu_{\text{back}}N_{\text{back}} = 0.80F_p$. Vertical equilibrium requires $f_{s,\text{shoes}} + f_{s,\text{back}} = mg$.

Hint 73 | Problem 73 – Accident Investigation on an Inclined Roadway

Acceleration down the incline is $a = g(\sin\theta - \mu_k\cos\theta)$. Use the kinematic relation $v^2 = v_0^2 + 2ad$ to calculate impact velocity v.

Hint 74 | Problem 74 – Crate Pressed Against a Truck

The normal force exerted on the crate by the accelerating truck is $N = ma$. For the crate not to slide downward, static friction must balance gravity: $f_s = mg \le \mu N = \mu ma$.

Hint 75 | Problem 75 – Stacked Block and Slab System

Find the maximum acceleration the block can sustain without slipping: $a_{\max} = \mu_s g$. Compare with the common acceleration of the combined system $a_{\text{comb}} = \frac{F}{M+m}$. If $a_{\text{comb}} > a_{\max}$, relative sliding occurs.

Hint 76 | Problem 76 – Tabletop Incline Projectile Launch

Ramp speed at exit is $v_0 = \sqrt{2gh}$. The block launches at angle $\theta = 30.0^\circ$ below the horizontal. Treat projectile motion using vertical displacement $-H = (-v_0\sin\theta)t - \frac{1}{2}gt^2$.

Hint 77 | Problem 77 – Accelerometer on an Inclined Cart

In the non-inertial frame of the accelerating cart, pseudo force $-m\vec{a}$ acts on the pendulum ball. Analyze components parallel and perpendicular to the incline to link string deflection angle $\theta/3$ to acceleration $a = g(\sin\theta - \mu\cos\theta)$.

Hint 78 | Problem 78 – Friction-Coupled Rod and Sliding Ball

Write Newton's second law for both bodies: $Mg - f = Ma_{\text{rod}}$ and $f - mg = ma_{\text{ball}}$. The relative acceleration is $a_{\text{rel}} = a_{\text{rod}} + a_{\text{ball}}$. Set $\frac{1}{2}a_{\text{rel}}(\Delta t)^2 = L$ to solve for f.

Hint 79 | Problem 79 – Time to Traverse Hanging Rod

Determine individual upward and downward accelerations from the pulley configuration constraint and Newton's second law. Find relative acceleration and set $s_{\text{rel}} = \frac{1}{2} a_{\text{rel}} t^2 = L$.

Hint 80 | Problem 80 – Maximum Ascent of Pulley System Mass

The motion consists of two phases: an accelerated phase while m_1 descends until it hits the floor, followed by a free-fall (upward projectile) phase for m_2 under gravity alone.